Addendum for Chapter 3
Curve Sketching and Real World Applications

MATH1123 Calculus for Science and Engineering

20 October 2026

Curve Sketching

Example 1

Sketch the graph of the function

$$f(x) = x^4 - 4x^3.$$

Indicate the points of local minima/local maxima, points of inflection, intervals of decrease/increase, intervals of concave up/concave down, and x- and y-intercepts.

Solution

Step 1. Find $f'(x)$, $f''(x)$, and the critical points.

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3),$$
$$f''(x) = 12x^2 - 24x = 12x(x-2).$$

$f'(x) = 0$: We have $4x^2(x-3) = 0$, so $x = 0, 3$ are critical points.
$f''(x) = 0$: We have $12x(x-2) = 0$, so $x = 0, 2$.

Step 2. Determine the signs of $f'(x)$ and $f''(x)$ as x increases.

x	$\rightarrow$	0	$\rightarrow$	3	$\rightarrow$
$4x^2$	$+$	0	$+$	$+$	$+$
$x-3$	$-$	$-$	$-$	0	$+$
Sign of $f'(x)$	$-$	0	$-$	0	$+$
Direction of $f(x)$	$\searrow$	$\rightarrow$	$\searrow$	$\rightarrow$	$\nearrow$

$f(x)$ is decreasing on $(-\infty, 3]$, and increasing on $[3, \infty)$.
Point of inflection at $x = 0$. $f(0) = 0$.
Local minimum at $x = 3$. $f(3) = 3^4 - 4(3^3) = -27 =$ value of local minimum.

x	$\rightarrow$	0	$\rightarrow$	2	$\rightarrow$
$12x$	$-$	0	$+$	$+$	$+$
$x-2$	$-$	$-$	$-$	0	$+$
Sign of $f''(x)$	$+$	0	$-$	0	$+$

$f(x)$ is concave up on $(-\infty, 0]$ and $[2, \infty)$, and concave down on $[0, 2]$.
Points of inflection at $x = 0, 2$. $f(0) = 0$, $f(2) = 2^4 - 4(2^3)$ $= -16$.

Step 3. Find x- and y-intercepts.

x-intercepts: Solve $f(x) = 0$. We have $x^4 - 4x^3 = 0$, so $x^3(x-4) = 0$, so $x = 0, 4$ are x-intercepts.

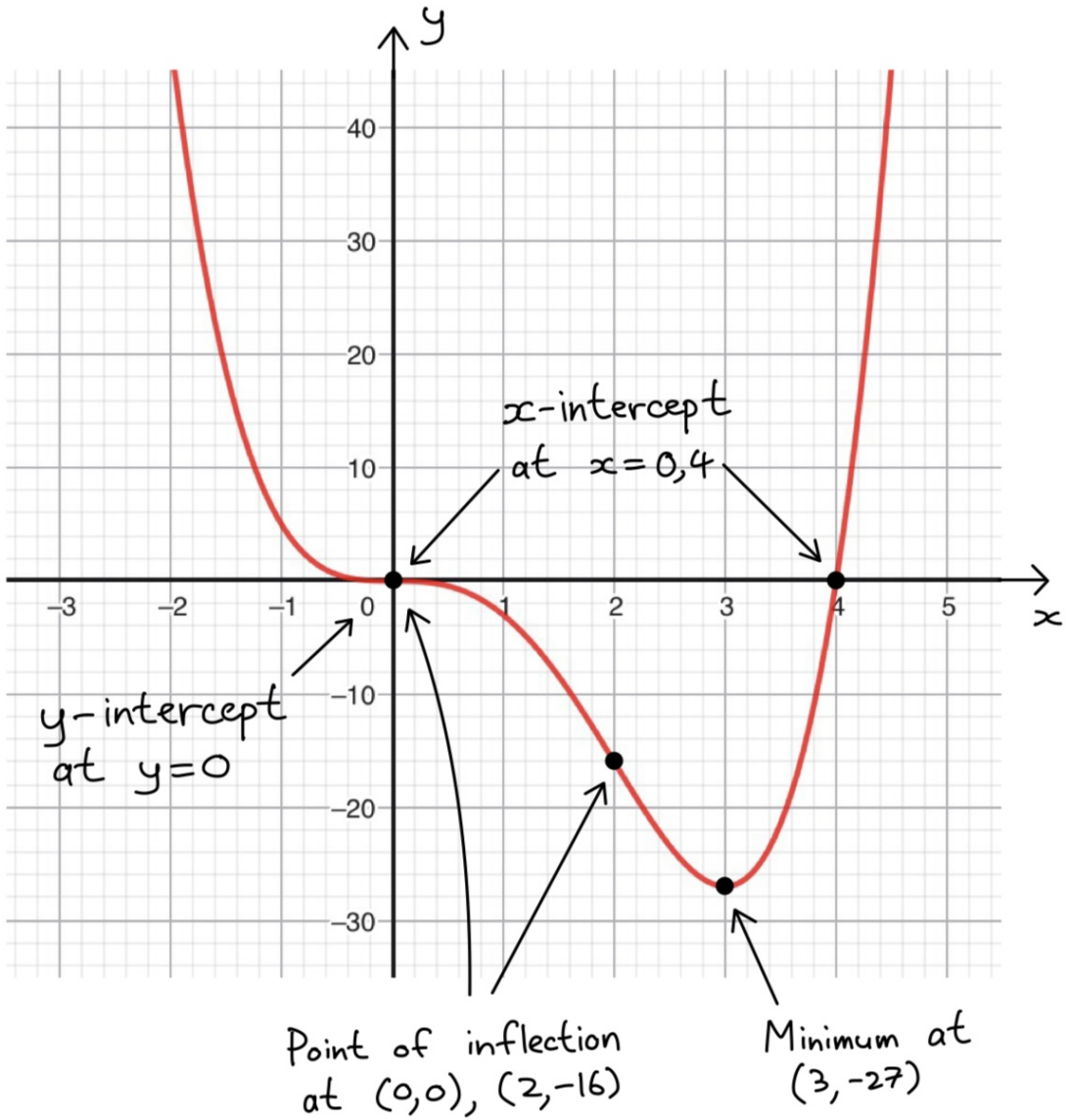
y-intercept: Find $f(0)$. We have $f(0) = 0$, so y-intercept is 0.

Step 4. Behaviour as $x \to \pm\infty$.

As $x \to \infty$, $f(x) \to \infty$.

As $x \to -\infty$, $f(x) \to \infty$.

Sketch of the graph of $f(x) = x^4 - 4x^3$.



Example 2

Sketch the graph of the function

$$f(x) = \frac{3x^2 - 4}{x^2 - 4}.$$

Indicate the points of local minima/local maxima, points of inflection, intervals of decrease/increase, intervals of concave up/concave down, asymptotes, and x- and y-intercepts.

Solution

Step 1. Find $f'(x)$, $f''(x)$, and the critical points.

$$f'(x) = \frac{6x(x^2-4)-(3x^2-4)(2x)}{(x^2-4)^2} = -\frac{16x}{(x^2-4)^2}$$
$$= -\frac{16x}{(x+2)^2(x-2)^2},$$

$$f''(x) = \frac{(-16)(x^2-4)^2+16x\cdot 2(x^2-4)\cdot 2x}{(x^2-4)^4} = \frac{16(3x^2+4)}{(x+2)^3(x-2)^3}.$$

$f'(x) = 0$ or $f'(x)$ undefined: We have $16x = 0$, so $x = 0$; and $(x+2)^2(x-2)^2 = 0$, so $x = -2, 2$. Critical points are $x = -2, 0, 2$.
$f''(x) = 0$ or $f''(x)$ undefined: $f''(x) = 0$ cannot happen since $16(3x^2+4) \geq 64 > 0$. $f''(x)$ undefined at $x = -2, 2$.

Step 2. Determine the signs of $f'(x)$ and $f''(x)$ as x increases.

x	$\rightarrow$	-2	$\rightarrow$	0	$\rightarrow$	2	$\rightarrow$
$-16x$	$+$	$+$	$+$	0	$-$	$-$	$-$
$(x+2)^2$	$+$	0	$+$	$+$	$+$	$+$	$+$
$(x-2)^2$	$+$	$+$	$+$	$+$	$+$	0	$+$
Sign of $f'(x)$	$+$	u.d.	$+$	0	$-$	u.d.	$-$
Direction of $f(x)$	$\nearrow$	$\updownarrow$	$\nearrow$	$\rightarrow$	$\searrow$	$\updownarrow$	$\searrow$

"u.d." means "undefined".
$f(x)$ is increasing on $(-\infty, -2)$ and $(-2, 0]$, and decreasing on $[0, 2)$ and $(2, \infty)$.
Local maximum at $x = 0$. $f(0) = \frac{0-4}{0-4} = 1 =$ value of local maximum.
Vertical asymptotes at $x = -2$ and $x = 2$.

x	$\rightarrow$	-2	$\rightarrow$	2	$\rightarrow$
$16(3x^2+4)$	$+$	$+$	$+$	$+$	$+$
$(x+2)^3$	$-$	0	$+$	$+$	$+$
$(x-2)^3$	$-$	$-$	$-$	0	$+$
Sign of $f''(x)$	$+$	u.d.	$-$	u.d.	$+$

$f(x)$ is concave up on $(-\infty,-2)$ and $(2,\infty)$, and concave down on $(-2,2)$.

Note: $x=-2,2$ are not points of inflection, since $f(x)$ is not continuous at $x=-2,2$, even though we have change between concave up and concave down.

Step 3. Find x- and y-intercepts.

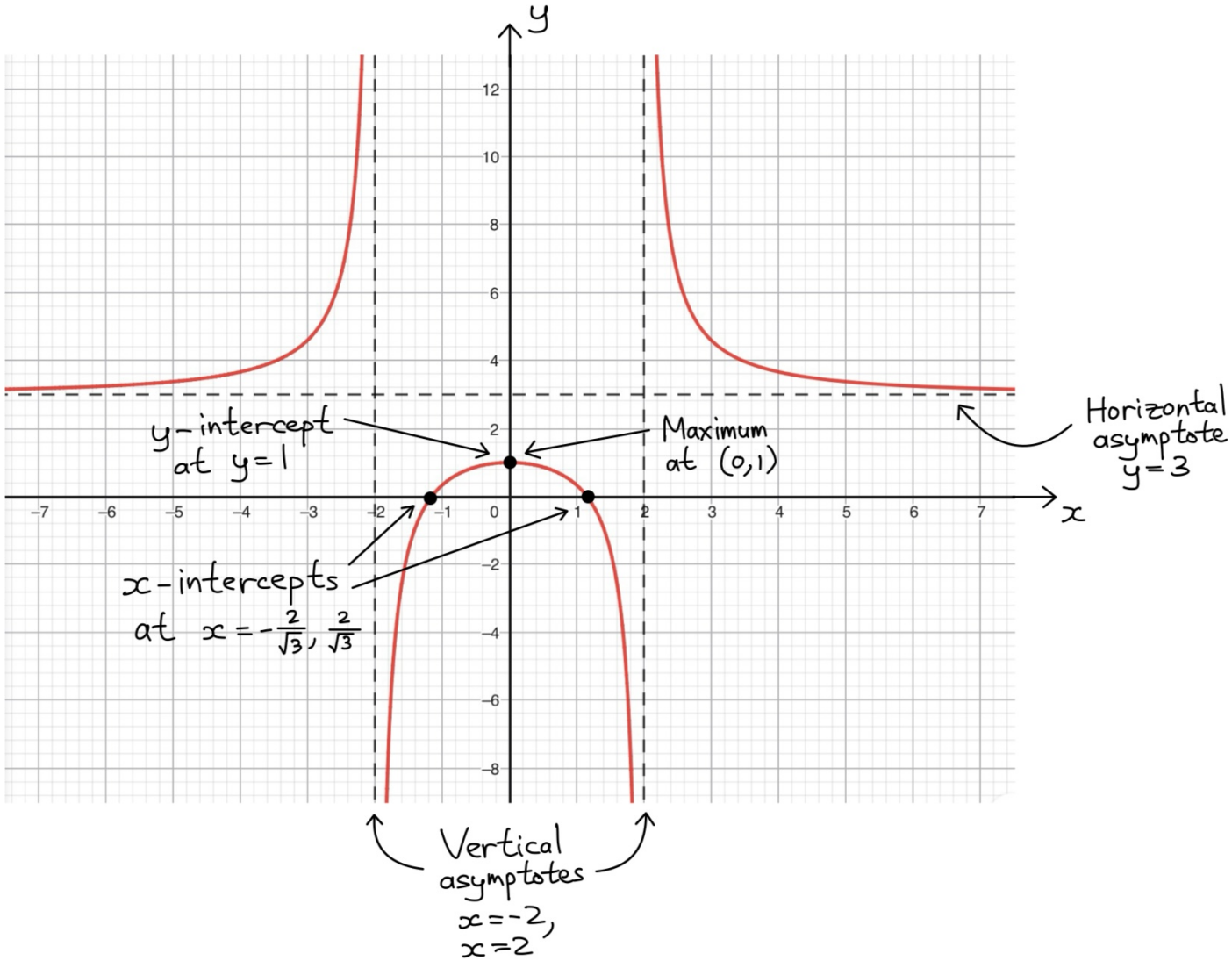
x-intercepts: Solve $f(x) = 0$. We have $3x^2 - 4 = 0$, so $x = -\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}$ are x-intercepts.

y-intercept: We have $f(0) = \frac{0-4}{0-4} = 1$, so y-intercept is 1.

Step 4. Behaviour as $x \to \pm\infty$.

As $x \to \infty$, $f(x) \to 3$.
As $x \to -\infty$, $f(x) \to 3$.
Horizontal asymptote is $y = 3$.

Sketch of the graph of $f(x) = \dfrac{3x^2 - 4}{x^2 - 4}$.



Real World Applications

Example 3

Maximising area. *A lifeguard needs to rope off a rectangular swimming area in front of a beach, using 180 metres of rope and floats. What dimensions of the rectangle will maximise the area? What is the maximum area? (Note that the beach is one side of the rectangle.)*



Solution

Let x be the side of the rectangle as indicated. The length of the rectangle parallel to the beach is $180 - 2x$. Hence, the area of the rectangle is $A(x) = x(180 - 2x) = 180x - 2x^2$. Note that we have $0 \leq x \leq 90$, which is the domain of the function $A(x)$. We determine the global maximum of $A(x)$ over $x \in [0, 90]$.

We have $A'(x) = 180 - 4x$. Critical points: Solve $A'(x) = 0$, so $180 - 4x = 0$, so $x = 45$ is the only critical point. Also, we have $A'(x) > 0$ for $0 \leq x < 45$, and $A'(x) < 0$ for $45 < x \leq 90$. By First Derivative Test, $A(x)$ has a local maximum at $x = 45$.

For the global extrema, we find $A(x)$ at the critical point $x = 45$, and at the end-points of the closed interval $[0, 90]$. We have

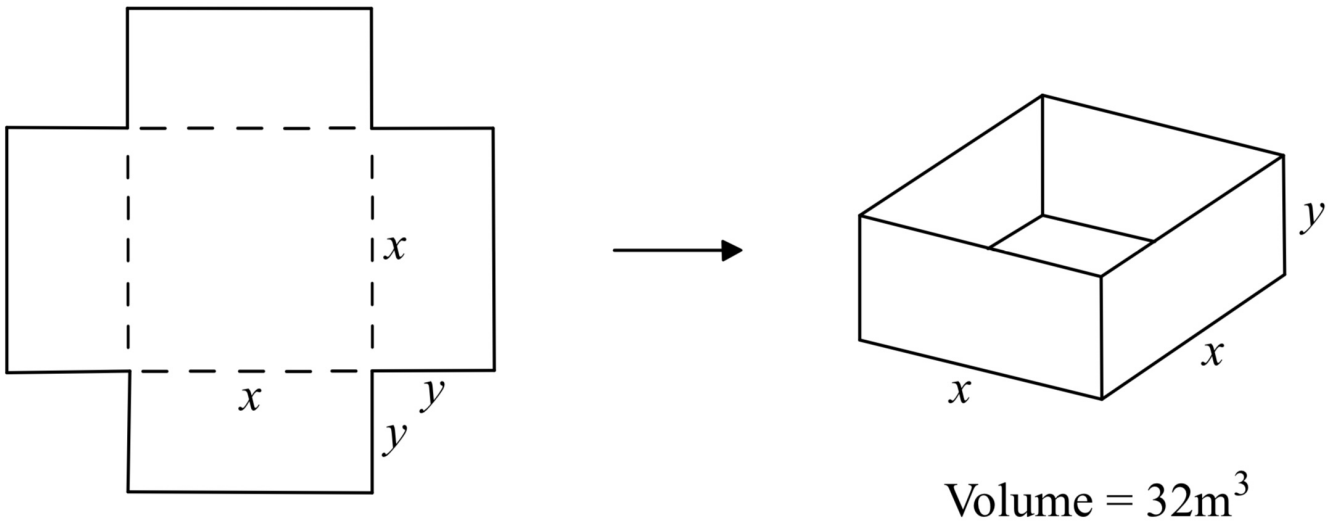
$$\begin{aligned} A(45) &= 45(180 - 2 \cdot 45) = 4050, \\ A(0) &= 0(180 - 2 \cdot 0) = 0, \\ A(90) &= 90(180 - 2 \cdot 90) = 0. \end{aligned}$$

We have the global maximum of 4050 for $A(x)$ at $x = 45$, and the global minimum of 0 at $x = 0, 90$. For $x = 45$, the length of the rectangle is $180 - 2 \cdot 45 = 90$.

We conclude that the maximum area of the rectangle is 4050m^2, and the dimensions achieving this maximum are 45m and 90m.

Example 4

Minimising surface area. *A water tank with the shape of an open-top, square-based, rectangular metal box is to be constructed so that it can hold a volume of* $32m^3$ *What dimensions will minimise surface area? What is the minimum surface area?*



Solution

Let x be the side length of the square base of the box, and y be the height of the box, as indicated. The volume of the box is x^2y, and this is required to be equal to 32, so $x^2y = 32$. The surface area of the box is $x^2 + 4xy = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + 128x^{-1}$. We shall find the global minimum of $S(x) = x^2 + 128x^{-1}$ over $x \in (0, \infty)$.

We have $S'(x) = 2x - 128x^{-2}$. Critical points: Solve $S'(x) = 0$, so $2x - 128x^{-2} = 0$, so $x^3 - 64 = 0$, so $x = 4$ is the only critical point. Also, $S'(x) < 0$ for $0 < x < 4$, and $S'(x) > 0$ for $x > 4$. By First Derivative Test, $S(x)$ has a local minimum at $x = 4$.

Moreover, by Increasing/Decreasing Function Theorem, we have $S(x)$ is decreasing on $(0, 4]$, and increasing on $[4, \infty)$. Hence, $S(x)$ has the global minimum of $S(4) = 4^2 + \frac{128}{4} = 48$ at $x = 4$. For $x = 4$, we have $y = \frac{32}{4^2} = 2$.

We conclude that the minimum surface area of the box is 48m^2, and the dimensions achieving this minimum are 4m for the side of the square base, and 2m for the height.